CS 340 Project Two: Grazioso Salvare Rescue Dashboard

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CS 340: Client/Server Development

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Project Overview

For this project, I created an interactive dashboard for Grazioso Salvare, a company that trains dogs for search and rescue work. The goal was to make it easier for the company to find dogs that meet certain requirements for different types of rescue training. I used MongoDB, Python, and Dash to create a dashboard where users can filter through animal shelter records and view the results in a table, bar chart, and map. This allows the company to look through the information more easily and identify dogs that could be good candidates for training.

Tools and Technologies Used

For this project, I used MongoDB, Python, Dash, Pandas, Plotly, and Dash Leaflet to build my dashboard. I used MongoDB to store the animal shelter data because it works well with Python and makes it easier to search for specific information. I also used the CRUD Python module I created in Project One to connect to the database and retrieve the records needed for each rescue

filter. Dash allowed me to make the dashboard interactive so that the table and charts update when a user selects a different rescue type. Pandas helped organize the data, Plotly was used to create the bar chart, and Dash Leaflet was used to show the animals' locations on a map. Together, these tools helped me create a dashboard that is easier for users to navigate and understand.

Dashboard Features and Functionality

For my dashboard, I added different features to make it easier for users to find and look through the animal shelter records. There are four options to choose from: Water Rescue, Mountain or Wilderness Rescue, Disaster or Individual Tracking, and Reset. When someone selects a rescue type, the table updates to show the dogs that match the requirements for that type of rescue work. The bar chart also changes to show the different breeds in the results. I also added a map that shows where an animal is located, and users can click on the marker to see the animal’s name. If someone wants to start over, they can select Reset, which brings back all the original records and updates the charts.

Dashboard Testing and Screenshots

After finishing my dashboard, I tested each of the filtering options to make sure everything worked correctly. I checked that the table and charts were updated when I selected different rescue types and that the Reset option brought back the original data. I also tested the map to make sure it displayed the selected animals' location.

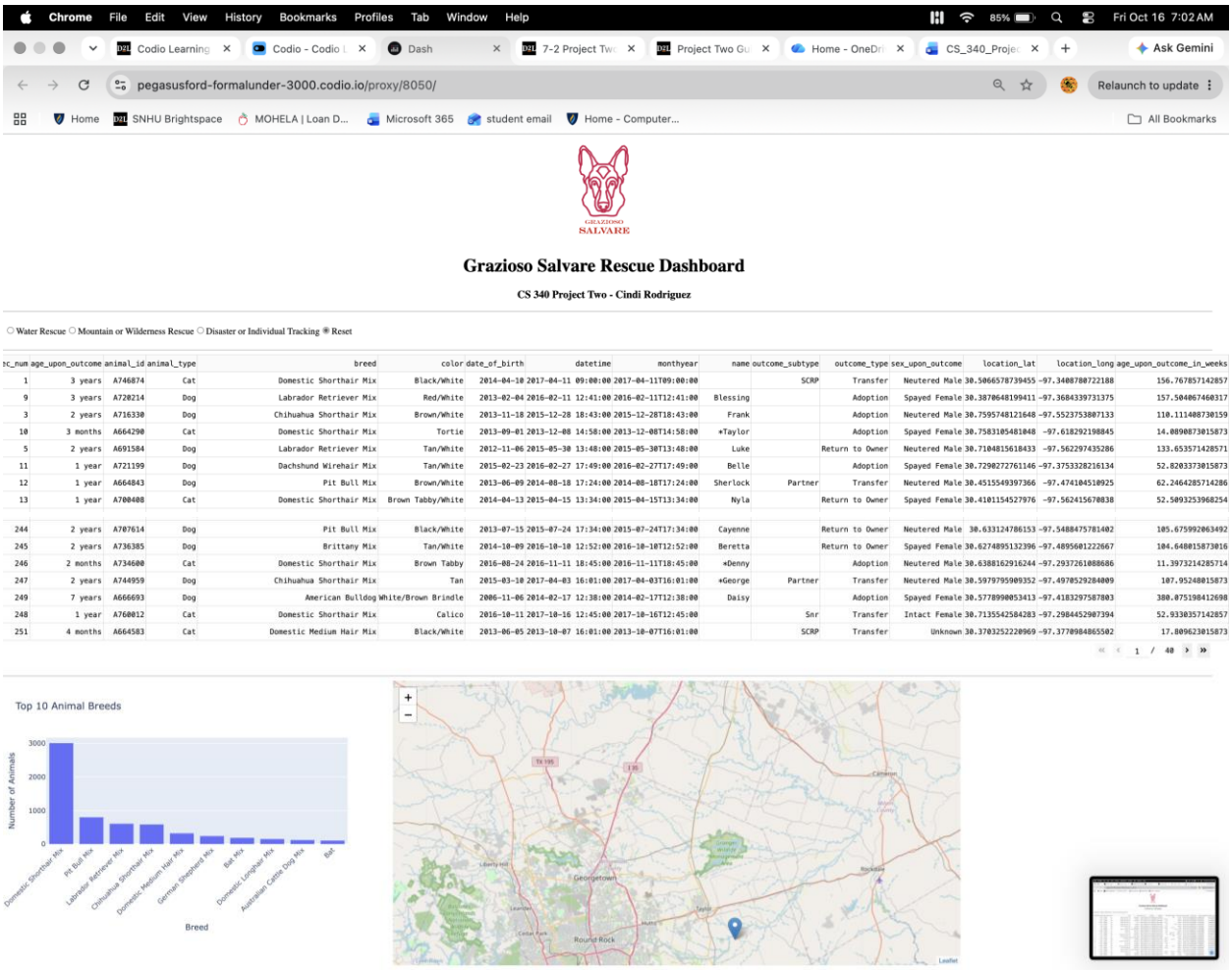


Figure 1: Original Dashboard (Unfiltered)

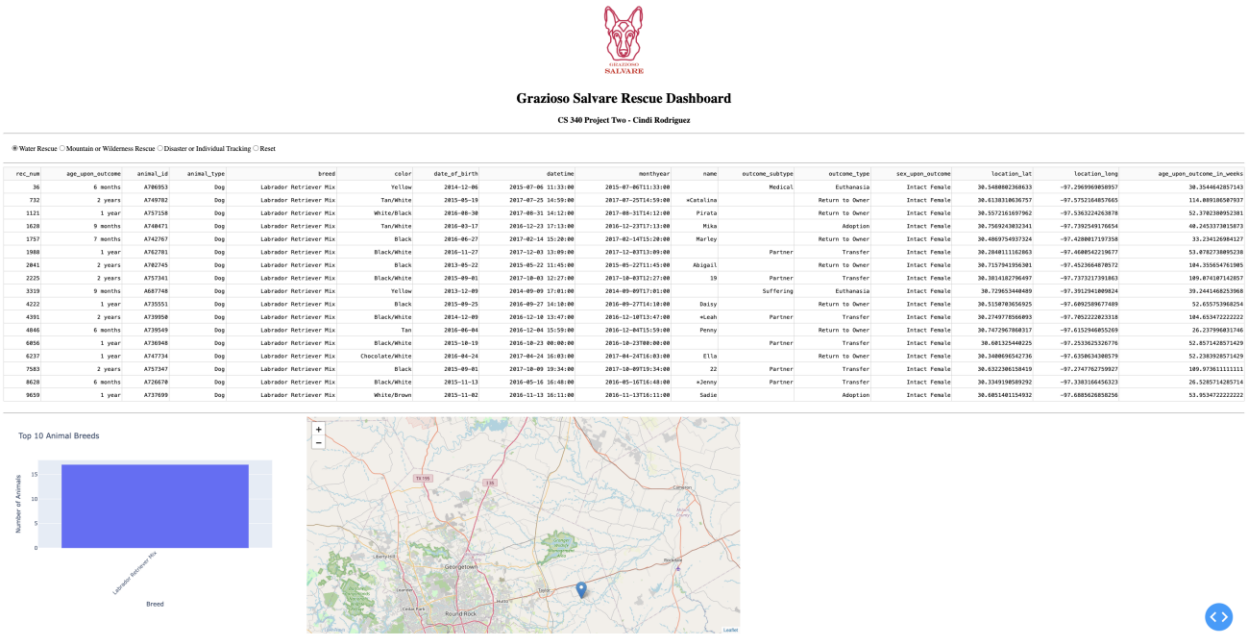


Figure 2: Water Rescue Filter

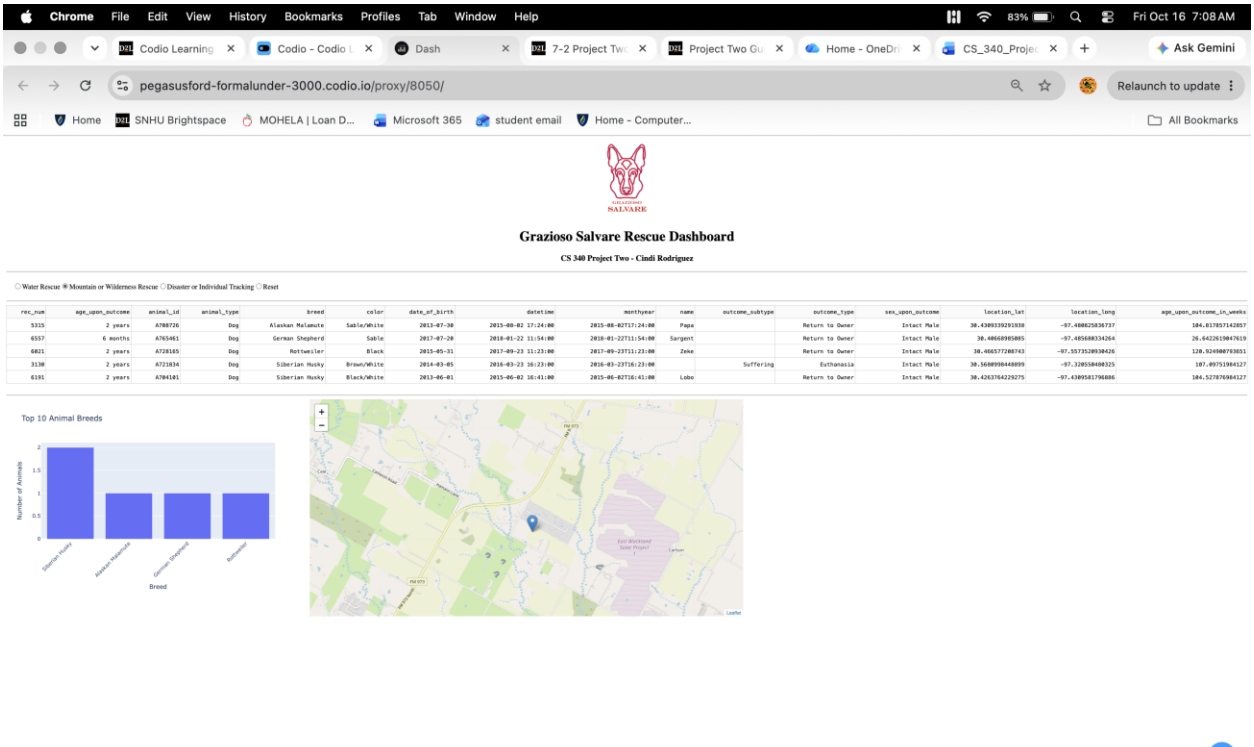


Figure 3: Mountain or Wilderness Rescue Filter

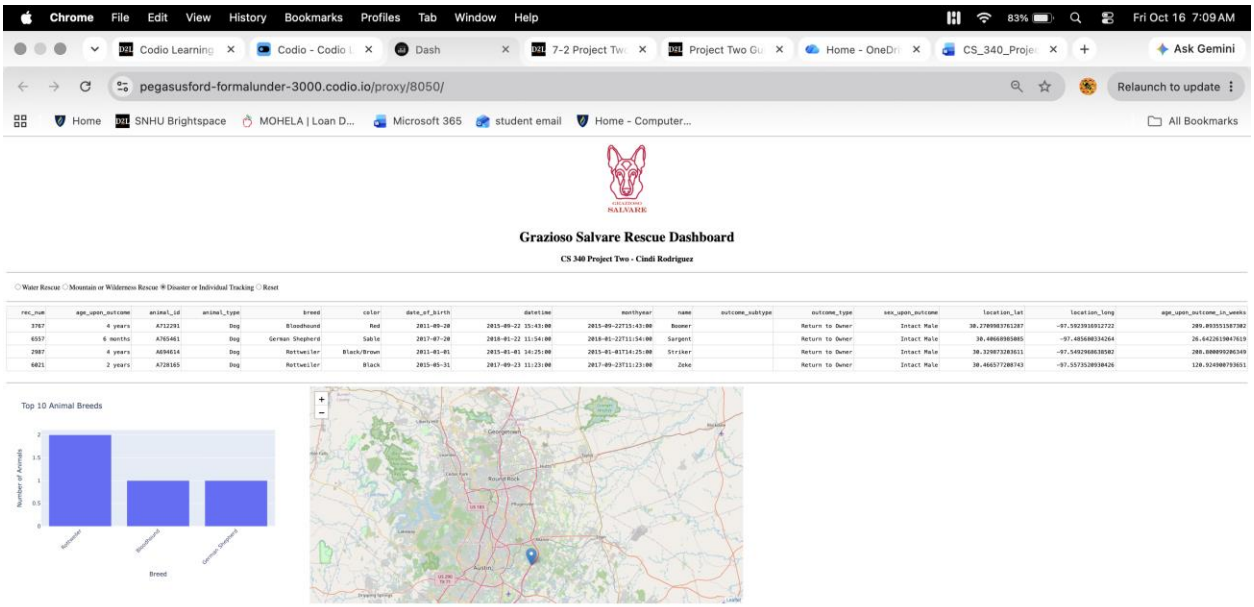
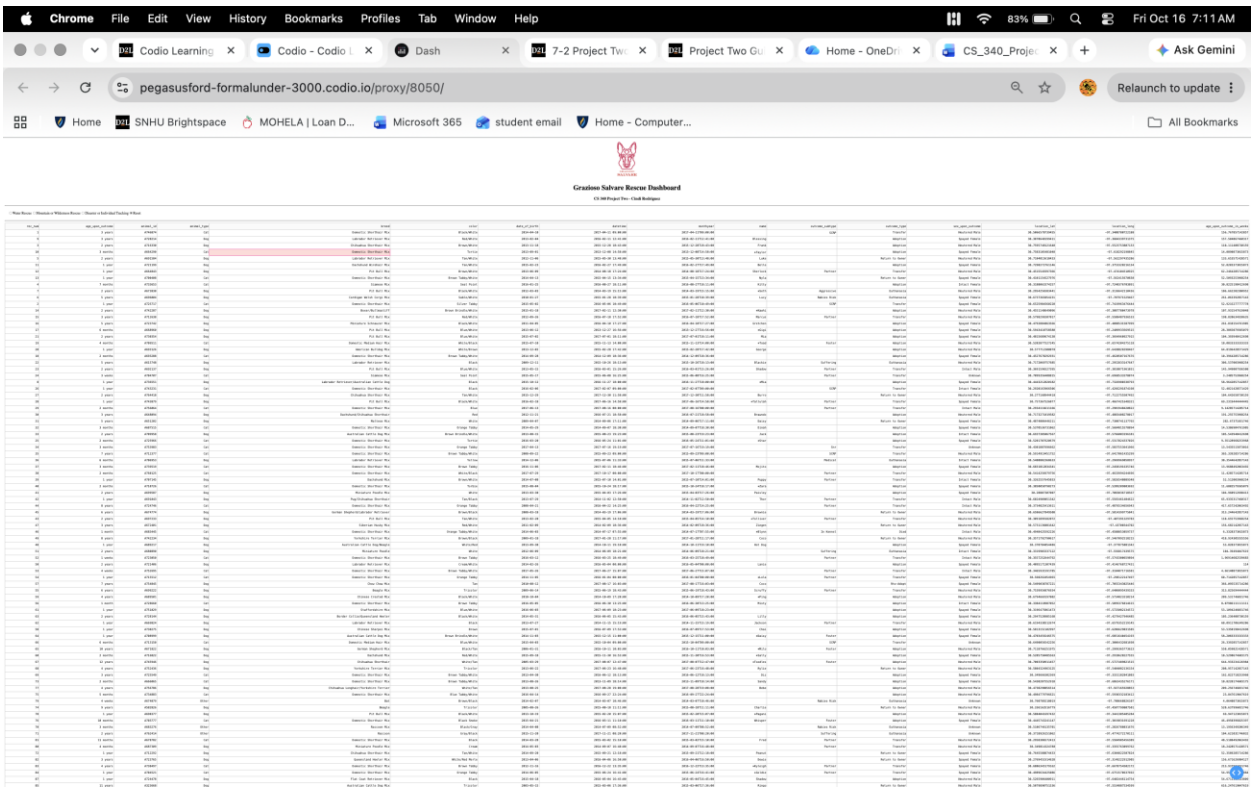


Figure 4: Disaster or Individual Tracker Filter



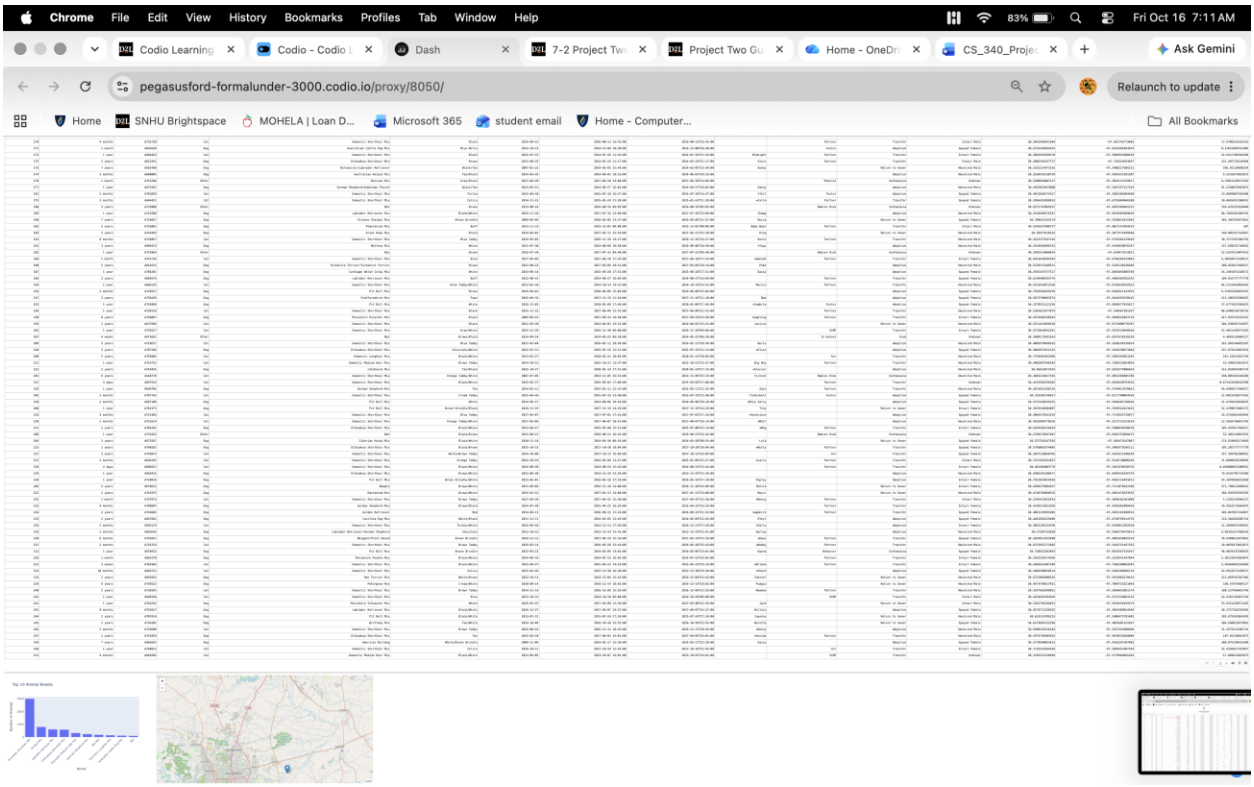


Figure 5: Reset Filter (Unfiltered Dashboard)

Steps to Reproduce the Project

To recreate this project, I first used MongoDB and the Austin Animal Center Outcomes dataset that was provided for the assignment. I used the CRUD Python module I created in Project One to connect to the database and pull the animal records. From there, I opened the ProjectTwoDashboard.ipynb file in Jupyter Lab through Codio and added the libraries I needed. I started with the data table and then added the different rescue filters so users could choose which type of dogs they wanted to see. Next, I added the bar chart and map and made sure they updated whenever a different filter was selected. After putting everything together, I ran the dashboard and tested each option to make sure everything was working the way it should.

Challenges and Solutions

One of the challenges I had with this project was getting everything to work together without running into errors. I had some trouble with the map because it would show an error when an animal wasn't selected. I fixed this by changing my code so it would check for missing data or selections before trying to display anything. I also had an issue with the table highlighting, which i fixed by adding a check for when no columns were selected. After fixing these issues, I went back and tested all the rescue filters to make sure the table, bar chart, and map were updating correctly. This helped me better understand how callbacks work and why it's important to test everything as I go.

Resources and References

MongoDB. (n.d.). *MongoDB documentation*. https://www.mongodb.com/docs

Python Software Foundation. (n.d.). *Python documentation*. https://docs.python.org/3/

Plotly. (n.d.). *Dash documentation*. https://dash.plotly.com/

pandas. (n.d.). *pandas documentation*. https://pandas.pydata.org/docs/

Austin Animal Center. (2020). *Austin Animal Center Outcomes* [Data set]. City of Austin, Texas Open Data Portal. https://doi.org/10.26000/025.000001